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Smart power selection for multi-electrode RF ablation system

Abstract

A computing device for generating and using a graphical user interface (GUI) is disclosed. The computing device includes one or more controllers configured to generate a graphical representation of a plurality of electrodes of an ablation catheter for displaying via the GUI; designate, via the GUI, at least some of the plurality of electrodes to be active electrodes; automatically designate the active electrodes as a source electrode or a sink electrode; assign an amount of energy to each of the designated source electrodes; and estimate an amount of energy associated with each of the designated sink electrodes based at least in part on the assigned energy of the designated source electrodes.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a Continuation application that claims priority to U.S. application Ser. No. 16/774,068, filed Jan. 28, 2020, now

TECHNICAL FIELD

(1) The present disclosure relates generally to systems, devices, and methods involving cardiac ablation.

BACKGROUND

(2) Various cardiac abnormalities can be attributed to improper electrical activity of cardiac tissue. Such improper electrical activity can include generation of electrical signals, conduction of electrical signals of the tissue, etc., in a manner that does not support efficient and/or effective cardiac function. For example, an area of cardiac tissue may become electrically active prematurely or otherwise out of synchrony during the cardiac cycle, causing the cardiac cells of the area and/or adjacent areas to contract out of rhythm. The result is an abnormal cardiac contraction that is not timed for optimal cardiac output. In some cases, an area of cardiac tissue may provide a faulty electrical pathway (e.g., a short circuit) that causes an arrhythmia, such as atrial fibrillation or supraventricular tachycardia. In some cases, inactive tissue (e.g., scar tissue) may be preferable to malfunctioning cardiac tissue.

SUMMARY

- (3) In Example 1, a computer-implemented method for displaying and using a graphical user interface (GUI) is disclosed. The method includes displaying, via the GUI, a graphical representation of a plurality of electrodes of an ablation catheter; designating, via the GUI, at least some of the plurality of electrodes to be active electrodes; designating the active electrodes as a source electrode or a sink electrode; assigning an electrical unit to each of the designated source electrodes; and estimating an amount of electrical units associated with each of the designated sink electrodes based at least in part on the assigned electrical unit of the designated source electrodes.
- (4) In Example 2, the method of Example 1, wherein estimating the electrical units associated with each of the designated sink electrodes is further based at least in part on distances between the designated source electrodes and the designated sink electrodes.
- (5) In Example 3, the method of Example 2, wherein the distances between the designated source electrodes and the designated sink electrodes represent edge-to-edge distances.
- (6) In Example 4, the method of Examples 2 or 3, wherein the distances are selected from a table of predetermined distances or are computed by one or more controllers.
- (7) In Example 5, the method of any of Examples 1-4, wherein estimating the electrical units associated with each of the designated sink electrodes includes adding each source's contribution to sink energy associated with each of the designated sink electrodes.
- (8) In Example 6, the method of any of Examples 1-5, further comprising modifying the assigned electrical units to at least one of the designated source electrodes or designating an additional active electrode or inactive electrode; and re-estimating the amount of electrical units associated with each of the designated sink electrodes based at least in part on the modified assigned electrical units of the at least one designated source electrode or the additional active or inactive electrode.
- (9) In Example 7, the method of any of Examples 1-6, further comprising determining that the estimated electrical units associated with at least one of the designated sink electrodes is above a threshold; and disabling an ablation-initiation button on the GUI in response to the determination.
- (10) In Example 8, the method of any of Examples 6 and 7, further comprising after re-estimating the electrical units associated with each of the designated sink electrodes, determining that the estimated electrical units associated with at least one of the designated sink electrodes is below the threshold; and enabling the ablation-initiation button on the GUI in response to determining that the estimated electrical units associated with at least one of the designated sink electrodes is below the threshold.
- (11) In Example 9, the method of any of Examples 1-8, further comprising displaying, via the GUI,

the estimated electrical units associated with each of the designated sink electrodes; and displaying, via the GUI, an indicator representing the estimated electrical unit being above or below one or more thresholds.

(12) In Example 10, the method of any of Examples 1-9, further comprising automatically modifying the assigned electrical units to each of the designated source electrodes in response to the estimated electrical units associated with each of the designated sink electrodes.

(13) In Example 11, the method of any of Examples 1-10, further comprising communicating the assigned electrical units or modified assigned electrical units for each of the designated source electrodes to a radio frequency generator.

(14) In Example 12, the method of any of Examples 1-11, wherein the electrical unit is power.

(15) In Example 13, a computing device adapted to execute the steps of the method of Examples 1-12.

(16) In Example 14, a computer program product comprising instructions to cause one or more processors to carry out the steps of the method of Examples 1-12.

(17) In Example 15, a computer-readable medium having stored thereon the computer program product of Example 14.

(18) In Example 16, an ablation system includes a radiofrequency (RF) generator configured to generate RF energy and an ablation catheter in communication with the RF generator and including a plurality of electrodes. The ablation system further includes one or more controllers in communication with the RF generator and configured to: generate a graphical representation of the plurality of electrodes of the ablation catheter for displaying via a graphical user interface (GUI); designate, via the GUI, at least some of the plurality of electrodes to be active electrodes; designate the active electrodes as a source electrode or a sink electrode; assign a power to each of the designated source electrodes; and estimate an amount of power associated with each of the designated sink electrodes based at least in part on the assigned power of the designated source electrodes.

(19) In Example 17, the ablation system of Example 16, wherein the estimated amount of power associated with each of the designated sink electrodes is further based at least in part on distances between the designated source electrodes and the designated sink electrodes.

(20) In Example 18, the ablation system of Example 17, wherein the distances between the designated source electrodes and the designated sink electrodes represent edge-to-edge distances.

(21) In Example 19, the ablation system of any of Examples 17 and 18, wherein the distances are selected from a table of predetermined distances stored in the one or more controllers or computed by the one or more controllers.

(22) In Example 20, the ablation system of any of Examples 16-19, wherein the estimated the amount of power associated with each of the designated sink electrodes is based at least in part on adding each source's contribution to power associated with each of the designated sink electrodes.

(23) In Example 21, the ablation system of any of Examples 16-20, wherein the one or more controllers is configured to: modify the assigned power to at least one of the designated source electrodes or designating an additional electrode as an active electrode or inactive electrode; and re-estimate the amount of power associated with each of the designated sink electrodes based at least in part on the modified assigned power of the at least one designated source electrode or the additional active electrode or inactive electrode.

(24) In Example 22, the ablation system of any of Examples 16-21, wherein the one or more controllers is configured to determine that the estimated power associated with at least one of the designated sink electrodes is above a threshold; and disable an ablation-initiation button on the GUI in response to the determination.

(25) In Example 23, the ablation system of any of Examples 21 and 22, wherein the one or more controllers is configured to after re-estimating the amount of power associated with each of the designated sink electrodes, determine that the estimated power associated with at least one of the

designated sink electrodes is below the threshold; and enable the ablation-initiation button on the GUI in response to determining that the estimated power associated with at least one of the designated sink electrodes is below the threshold.

(26) In Example 24, the ablation system of any of Examples 16-23, wherein the ablation catheter includes an expandable member carrying the plurality of electrodes.

(27) In Example 25, the ablation system of any of Examples 16-24, further comprising a display in communication with the one or more controllers and configured to display the GUI.

(28) In Example 26, a computing device for generating and using a graphical user interface (GUI) is disclosed. The computing device includes one or more controllers configured to: generate a graphical representation of a plurality of electrodes of an ablation catheter for displaying via the GUI; designate, via the GUI, at least some of the plurality of electrodes to be active electrodes; automatically designate the active electrodes as a source electrode or a sink electrode; assign an amount of energy to each of the designated source electrodes; and estimate an amount of energy associated with each of the designated sink electrodes based at least in part on the assigned energy of the designated source electrodes.

(29) In Example 27, the computing device of Example 26, wherein the estimated amount of energy associated with each of the designated sink electrodes is further based at least in part on distances between the designated source electrodes and the designated sink electrodes.

(30) In Example 28, the computing device of Example 27, wherein the distances between the designated source electrodes and the designated sink electrodes represent edge-to-edge distances.

(31) In Example 29, the computing device of Example 27, wherein the distances are selected from a table of predetermined distances stored in the one or more controllers.

(32) In Example 30, the computing device of Example 26, wherein the estimated the amount of energy associated with each of the designated sink electrodes is based at least in part on adding each source's contribution to energy associated with each of the designated sink electrodes.

(33) In Example 31, the computing device of any of Examples 26-30, wherein the one or more controllers includes a computer readable storage medium having program code stored thereon for execution by the one or more controllers to carry out the functions of Examples 26-30.

(34) In Example 32, the computing device of Example 26, wherein the one or more controllers is configured to determine that the estimated amount of energy associated with at least one of the designated sink electrodes is above a threshold; and disable an ablation-initiation button on the GUI in response to the determination.

(35) In Example 33, a computing device for generating and using a graphical user interface (GUI) is disclosed. The computing device includes one or more controllers configured to: generate a graphical representation of a plurality of electrodes of an ablation catheter for displaying via the GUI; designate, via the GUI, at least some of the plurality of electrodes to be active electrodes; automatically designate the active electrodes as a source electrode or a sink electrode; and automatically assign a power to each of the designated source electrodes such that an estimated power associated with each of the designated sink electrodes is less than a first predetermined threshold.

(36) In Example 34, the computing device of Example 33, wherein the estimated power of a first set of the sink electrodes is compared against the first predetermined threshold, wherein the estimated power of a second set of the sink electrodes is compared against a second predetermined threshold that is different than the first predetermined threshold.

(37) In Example 35, the computing device of Example 33, further comprising automatically assign a power to each of the designated source electrodes such that an estimated power associated with each of the designated sink electrodes is substantially similar to the estimated power of the other sink electrodes.

(38) While multiple embodiments are disclosed, still other embodiments of the present invention will become apparent to those skilled in the art from the following detailed description, which

shows and describes illustrative embodiments of the invention. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature and not restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows an ablation system, in accordance with certain embodiments of the present disclosure.
- (2) FIG. 2 shows a side, cut-away view of an ablation catheter, in accordance with certain embodiments of the present disclosure.
- (3) FIGS. 3-5 show various views of a graphical user interface, in accordance with certain embodiments of the present disclosure.
- (4) FIG. 6 shows an example power selection icon, in accordance with certain embodiments of the present disclosure.
- (5) FIG. 7 shows a conceptual schematic of source electrodes and sink electrodes, in accordance with certain embodiments of the present disclosure.
- (6) FIG. 8 shows a schematic of steps of estimating an amount of electrical units associated sink electrodes, in accordance with certain embodiments of the present disclosure.
- (7) While the invention is amenable to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and are described in detail below. The intention, however, is not to limit the invention to the particular embodiments described. On the contrary, the invention is intended to cover all modifications, equivalents, and alternatives falling within the scope of the invention as defined by the appended claims.

DETAILED DESCRIPTION

- (8) Cardiac ablation is a procedure by which cardiac tissue is treated to inactivate the tissue. The tissue targeted for ablation may be associated with improper electrical activity, as described above. Cardiac ablation can create lesions in the tissue and prevent the tissue from improperly generating or conducting electrical signals. For example, lesions in the form of a line or a circle can block the propagation of errant electrical signals. Control of the shape, depth, uniformity, etc., of the lesion is desirable.
- (9) Certain embodiments of the present disclosure involve systems, devices, and methods that can be used in connection with cardiac ablation. In particular, the present disclosure describes graphical user interfaces that display and enable control of a graphical representation of features of the ablation catheter. The graphical user interfaces can be used for viewing, selecting, and modifying ablation parameters, among other things. For example, the graphical user interfaces can be used to view, select, and modify ablation patterns and energy associated with ablation electrodes. As such, the graphical user interfaces can be used to help control the shape, depth, and uniformity of lesions.
- (10) FIG. 1 shows an ablation system **100** including an ablation catheter **102** comprising an elongated catheter body **104** and a distal catheter region **106**, which is configured to be positioned within a heart **108**. The ablation catheter **102** includes an expandable member **110** (e.g., membrane, balloon) and a plurality of energy delivery elements **112** (e.g., ablation electrodes) secured to the expandable member **110**. The energy delivery elements **112** are configured and positioned to deliver ablative energy (e.g., radiofrequency energy) to tissue when the expandable member **110** is inflated.
- (11) The system **100** includes a radiofrequency (RF) generator **114** electrically coupled to the plurality of energy delivery elements **112** and configured to generate RF energy. The RF generator **114** includes an RF generator controller **116** configured to control the RF energy to the plurality of energy delivery elements **112**. The RF generator controller **116** can be implemented using firmware, integrated circuits, and/or software modules that interact with each other or are combined

together. For example, the RF generator controller **116** may include memory **118** storing computer-readable instructions/code **120** for execution by a processor **122** (e.g., microprocessor) to perform aspects of embodiments of methods discussed herein.

(12) The system **100** can also include a computing device **124** (e.g., personal computer) with a display controller **126** configured to communicate with various components of the system **100** and generate a graphical user interface (GUI) to be displayed via a display **128** (e.g., computer monitor, television, mobile device screen). The display controller **126** can be implemented using firmware, integrated circuits, and/or software modules that interact with each other or are combined together. For example, the display controller **126** may include memory **130** storing computer-readable instructions/code **132** for execution by a processor **134** (e.g., microprocessor) to perform aspects of embodiments of methods discussed herein.

(13) The various components of the system **100** may be communicatively coupled to each other via communication links **136**. In certain embodiments, the communication links **136** may be, or include, a wired communication link (e.g., a serial communication), a wireless communication link such as, for example, a short-range radio link, such as Bluetooth, IEEE 802.11, a proprietary wireless protocol, and/or the like. The term “communication link” may refer to an ability to communicate some type of information in at least one direction between at least two components and may be a persistent communication link, an intermittent communication link, an ad-hoc communication link, and/or the like. The communication links **136** may refer to direct communications between components and/or indirect communications that travel between components via at least one other device (e.g., a repeater, router, hub).

(14) In embodiments, the memory **118** and **130** includes computer-readable storage media in the form of volatile and/or nonvolatile memory and may be removable, non-removable, or a combination thereof. Media examples include Random Access Memory (RAM), Read Only Memory (ROM), Electronically Erasable Programmable Read Only Memory (EEPROM), flash memory, and/or any other non-transitory storage medium that can be used to store information and can be accessed by a computing device. In certain embodiments, the ablation catheter **102** includes memory that stores information unique to the ablation catheter **102** (e.g., catheter ID, manufacturer). This information can be accessed and associated with data collected as part of an ablation procedure (e.g., patient data, ablation parameters).

(15) The computer-executable instructions **120** and **132** may include, for example, computer code, machine-useable instructions, and the like such as, for example, program components capable of being executed by the one or more processors **122** and **134**. Some or all of the functionality contemplated herein may be implemented in hardware and/or firmware.

(16) In certain embodiments, the RF generator **114** and the computing device **124** are separate components housed in a single console **138**. In certain embodiments, the RF generator **114** and the computing device **124** each include a plurality of controllers that are configured to perform aspects of embodiments of methods and procedures discussed herein.

(17) FIG. 2 shows an ablation catheter **200** that can be used in the system **100**. The ablation catheter **200** includes an expandable member **202** and a plurality of energy delivery elements **204** secured to the expandable member **202**. The energy delivery elements **204** are configured and positioned to deliver ablative energy to tissue when the expandable member **202** is inflated. As shown in FIG. 2, in certain embodiments, the energy delivery elements **204** are arranged in two rows, one proximal set of energy delivery elements **204** and one distal row of energy delivery elements **204**. Each of the energy delivery elements **204** is individually addressable or can be used with any other energy delivery element **204**. The energy delivery elements **204** can operate in a monopolar mode or bipolar mode. Sets of energy delivery elements **204** can be chosen such that the lesion is linear, a spot, a hollow circle, etc. In embodiments utilizing a monopolar mode, the system **100** may include a return pad.

(18) The ablation catheter **200** includes a visualization system **206** including a camera assembly

208 and illumination sources (e.g., light-emitting diodes (LEDs)) disposed on a guide wire shaft **210**. The camera assembly **208** can include a plurality of cameras disposed at an angle relative to a longitudinal axis of the ablation catheter **200**. The cameras are configured to enable real-time imaging (e.g., video) of an ablation procedure from within the expandable member **202** including visualizing the expandable member **202**, the energy delivery elements **204**, and cardiac tissue as well as lesion formation during the ablation procedure. The illumination sources provide lighting for the cameras to visualize the ablation procedure.

(19) As mentioned above, the computing device **124** of the system **100** includes the display controller **126** that is configured to communicate with various components of the system **100** and generate a GUI for displaying via the display **128**. FIGS. 3-5 show an example GUI and its various features and views that can be used in the system **100** and displayed via the display **128**. Users can interact (e.g., select icons, enter data) with the GUI using one or more input devices (e.g., mouse, keyboard, touchscreen). The various icons described below can take the form of selectable buttons, indicators, images, etc., on the GUI.

(20) FIGS. 3-5 show a GUI **300** including a first region **302** and a second region **304**. The first region **302** displays a graphical representation **306** of electrodes of an ablation catheter, and the second region **304** displays a real-time video **308** from the ablation catheter. The first region **302** and the second region **304** are shown as being positioned side-by-side and being circular-shaped regions. In certain embodiments, the first region **302** and the second region **304** are separate windows within the GUI **300**.

(21) The graphical representation **306** includes a separate electrode icon **310** for each of the plurality of electrodes of the ablation catheter. In certain embodiments, each electrode icon **310** is similarly-shaped to an actual shape of a corresponding electrode on the ablation catheter. Each electrode icon **310** can include a unique numerical indicator **312**. For example, in the embodiment shown in FIGS. 3-5, the ablation catheter being represented by the graphical representation **306** includes twelve electrodes in an outer ring and six electrodes in an inner ring, and each of the electrode icons **310** is assigned an integer (e.g., 1-18). A user can select or deselect an electrode icon **310** to respectively highlight or un-highlight the electrode icon **310** on the GUI **300**. As will be discussed in more detail below, a selected electrode icon **310** will be designated to be an active electrode (e.g., a source electrode or a sink electrode). The user can select electrode icons **310** for particular paths of ablation. FIG. 4 shows an example graphical representation **306** of electrode icons **310** some of which are selected (i.e., electrodes 1-6, 12, and 16-18) to be active electrodes and the rest of the electrode icons **310** unselected such that ablation electrodes associated with such electrode icons will not be active during an ablation procedure.

(22) The displayed real-time video **308** allows for visualization of an ablation procedure. The displayed real-time video **308** may include displaying video recorded by one or more cameras. For example, if an ablation catheter (e.g., the ablation catheter **200** of FIG. 2) includes four cameras, the real-time video **308** may display video recorded from each of the four cameras. In such embodiments, the real-time video **308** can display each of the four fields of view from the cameras overlaid with at least one other field of view.

(23) The GUI **300** includes a number of icons (e.g., buttons, images, combinations thereof) that are associated with and can be used to control or monitor aspects of the ablation catheter and the GUI **300** itself.

(24) FIG. 3 shows the GUI **300** including icons relating to the graphical representation **306** positioned in or near the first region **302** next to the graphical representation **306**. For example, the GUI **300** includes three icons (i.e., an electrode selection icon **314**, an electrode refresh icon **316**, and a source-sink reverse icon **318**) positioned next to the graphical representation **306** and that affect features of the ablation system **100**. The electrode selection icon **314** can be used to select a pattern from a pre-determined menu of patterns of electrode icon selections **310** (e.g., inner ring of electrode icons **310**, outer ring, all electrode icons **310**, none). Once a pattern is selected, the

selected electrode icons **310** can be highlighted on the GUI **300**. The electrode refresh icon **316** can be used to unselect any electrode icon **310** that has been selected. The source-sink reverse icon **318** can be used in a bipolar mode to reverse which electrode icons **310** correspond to a sink and which electrode icons **310** correspond to a source.

(25) FIG. **3** shows the GUI **300** including icons relating to the real-time video **308** positioned in or near the second region **304** next to the real-time video **308**. For example, the GUI **300** includes three icons (i.e., a contrast icon **320**, a luminosity icon **322**, and a video refresh icon **324**) positioned next to the real-time video **308** and that affect features of the real-time video **308**. The contrast icon **320** can be used to increase or decrease contrast of the real-time video **308**. The luminosity icon **322** can be used to modify illumination power of illumination sources in the ablation catheter. The video refresh icon **324** can be used to refresh the video feed and/or a display controller if the real-time video **308** encounters problems.

(26) The GUI **300** includes a ribbon **326** with various icons relating to the ablation catheter and/or the GUI **300** itself. The ribbon **326** includes a status icon **328** indicating the system's status and a sonic/scan icon **330**, which initiates a routine for initiating an ultrasonic source and for scanning the electrodes on the ablation catheter to identify potentially faulty electrodes. For example, the ablation catheter may be placed in a bath coupled to an ultrasonic source, and once the sonic/scan icon **330** is selected, the routine can turn on the ultrasonic source for a predetermined period of time to remove air bubbles stuck to the ablation catheter before a treatment procedure. After expiration of the predetermined period of time, the routine can sequentially activate all electrodes to determine whether any electrodes or RF amplifiers are defective. If the ultrasonic source is not connected, the sonic/scan icon **330** will just initiate the scanning portion of the routine.

(27) The ribbon **326** also includes an in vivo icon **332**, which can be selected to indicate that the ablation catheter has been placed within a patient; an anatomy icon **334**, which can be used to identify the pulmonary vein (e.g., right superior, right inferior, left superior, left inferior) to be treated; a power icon **336**, which displays and allows a user to modify, via arrow buttons, a power level at which the selected ablation electrodes will be energized; a procedure timing icon **338**, which displays and allows a user to modify, via arrow buttons, the length of time the selected ablation electrodes are to be energized; an irrigation flow rate icon **340**, which can be used to control flow rates of irrigation fluid through the ablation catheter; and a fluid volume icon **342**, which indicates the amount of fluid passed through the ablation catheter since the in vivo icon **332** was selected. Once various selections are made via the icons, data associated the selections can be stored in a memory (e.g., the memory **118** and **130** of FIG. **1**) and/or sent to a computing device (e.g., the computing device **124** of FIG. **1**). For example, once a flow rate is selected, the selected flow rate can be sent to the computing device to control an irrigation fluid pump.

(28) As described above, the GUI **300** allows a user to select, via the electrode icons **310**, which electrodes on the ablation catheter will be active (e.g., either a source electrode or a sink electrode). The selected and highlighted electrode icons **310** indicate that, should an ablation procedure begin, only the ablation electrodes corresponding to the selected and highlighted electrode icons **310** will be active during the ablation procedure. FIG. **3** shows the GUI **300** including an ablation activate/deactivate icon **344** with text stating "Assign" or one or more similar terms. Once the "Assign" ablation activate/deactivate icon **344** has been selected, the "Assign" text for the ablation activate/deactivate icon **344** is replaced with "Ablate", "Start", or one or more similar terms.

(29) When the ablation activate/deactivate icon **344** indicates "Assign" and is selected, a computing device assigns or designates the selected electrode icons **310** to be a source or a sink. For example, as shown in FIG. **4**, electrode icons **310** numbered 1-6, 12, and 16-18 are selected, and FIG. **5** shows the selected electrode icons **310** as being either a source electrode or a sink electrode. In FIG. **5**, the selected electrode icons **346** numbered "12", "2", "4", "6", and "17" are indicated as being source electrodes while the other selected electrode icons **348** (i.e., those numbered "1", "3", "5", "16", and "18") are indicated as being sink electrodes. In certain embodiments, the source

electrodes are highlighted with a different color on the GUI **300** or are otherwise shown as being different than the sink electrodes.

(30) FIG. **5** shows each of the source electrode icons **346** and the sink electrode icons **348** including icons indicating electrical units associated with the given electrode. For simplicity, the description below uses power in the form of Watts as the exemplary electrical unit displayed and modified via the GUI **300**, but other electrical units (e.g., various forms of electrical energy) can be displayed and modified in place of power.

(31) FIG. **5** shows the source electrode icons **346** including a source power icon **350**, which displays the power (e.g., 8 Watts) currently assigned to the corresponding ablation electrode. Each source power icon **350** can be selected to display a power selector icon **352**, an example of which is shown in FIG. **6**. The power selector icon **352** can cover the source power icon **350** on the GUI **300** until the user confirms or cancels changes in power.

(32) The power selector icon **352** includes icons such as a power-increase icon **354** with a “+” sign for increasing the power associated with the given ablation electrode, a power-decrease icon **356** with a “-” sign for decreasing the power associated with the given ablation electrode, a power-confirmation icon **358** with a check-mark for confirming changes in power, and a power-cancellation icon **360** with an “X” for cancelling changes in power. The amount of power for each ablation electrode can be modified in real time. As power is increased or decreased for a given ablation electrode, the electrode icons **310** can be shaded or brightened to visually indicate that such electrode icons **310** have a different power.

(33) FIG. **5** shows each of the sink electrodes **348** including a power estimation icon **362**. Each of the power estimation icons **362** displays an estimated power associated with the given sink electrode **348**. The estimated power displayed in the power estimation icons **362** is based at least in part on the power assigned to each of the source electrodes **346** and distances between the given sink electrode **348** and source electrodes **346**. For example, a source electrode **346** will divide its energy among the sink electrodes **348** but more of its energy will be delivered to sink electrodes **348** positioned closer to the given source electrode **346**.

(34) FIG. **7** shows a schematic **400** of electrodes that helps illustrate how energy from source electrodes is estimated to contribute to energy received at sink electrodes, and FIG. **8** shows a schematic **500** of steps for estimating an amount of electrical units (e.g., energy, power) associated with each of the designated sink electrodes. The steps outlined in FIG. **8** and described below can be performed sequentially or in parallel and in various orders.

(35) FIG. **8** shows a first step **502** of determining the number of source electrodes and the number of sink electrodes. The schematic **400** in FIG. **7** includes two source electrodes (i.e., a first source electrode **402A** and a second source electrode **402B**) and three sink electrodes (i.e., a first sink electrode **404A**, a second sink electrode **404B**, and a third sink electrode **404C**). After determining the number of source electrodes and sink electrodes, FIG. **8** shows a step **504** of determining the assigned energy (e.g., power) for each of the source electrodes. The first source electrode **402A** is associated with a power of 8 Watts, and the second source electrode **402B** is associated with a power of 6 Watts. Step **506** involves determining the distances between each source electrode and each sink electrode. As shown in FIG. **7**, the first source electrode **402A** is positioned 2 units away from the first sink electrode **404A**, 4 units from the second sink electrode **404B**, and 3 units from the third sink electrode **404C**. The second source electrode **402B** is positioned 6 units away from the first sink electrode **404A**, 4 units from the second sink electrode **404B**, and 1 unit from the third sink electrode **404C**.

(36) The estimated energy to be received by the first sink electrode **404A**, the second sink electrode **404B**, and the third sink electrode **404C** can be calculated based at least in part on the energy associated with each of the source electrodes and the distances between the source electrodes and the respective sink electrodes (step **508** in FIG. **8**). Equation 1 is one example approach for calculating the estimated energy to be received by the sink electrodes, where $I_{sub.S1}$ is the

estimated energy at the first sink **404A**, $L_{\text{sub.S1,Di}}$ =the distance between the sources and the first sink **404A**, $L_{\text{sub.S2,Di}}$ =the distance between the sources and the second sink **404B**, $L_{\text{sub.S3,Di}}$ =the distance between the sources and the third sink **404B**, and $I_{\text{sub.Di}}$ =the sources' respective assigned energy.

$$(37) \quad I_{S1} = \text{Math.} \frac{L_{S1, Di}^{-1}}{L_{S1, Di}^{-1} + L_{S2, Di}^{-1} + L_{S3, Di}^{-1}} * I_{Di} \quad \text{Equation 1}$$

(38) As shown in Equation 1, the estimated power for each sink electrode is a summation of each source electrode's contribution to power at the sink electrode. When each of the source electrodes is assigned the same amount of power, the source electrode positioned farthest from a given sink electrode will be estimated to contribute the least to the power of the given sink electrode.

(39) In certain embodiments, distances between source electrodes and sink electrodes are based on arc-length distances between edges (e.g., edge-to-edge distances) of the respective electrodes (e.g., energy delivery elements **204** of FIG. 2) when the expandable member **202** is substantially deployed (e.g., substantially fully expanded). FIG. 2 includes three examples of edge-to-edge distances (i.e., D1, D2, and D3) between electrodes. The edge-to-edge distances are arc-length distances to account for the curved shape of the expandable member **202** when fully deployed. In certain embodiments, the distances are predetermined distances stored in a table and accessed by the system. In other embodiments, the distances between source electrodes and sink electrodes are computed. For example, the electrodes may include position tracking features or be capable of sensing signals such that the distances between the electrodes can be computed in real time. In such embodiments, the geometry of the electrodes may be stored in a table and accessed by the system.

(40) In certain embodiments, the estimated power displayed in each of the power estimation icons **362** is updated as a user modifies power assigned for a given source electrode **346** or as more electrodes are designated as being inactive or active (and therefore a source electrode or a sink electrode).

(41) In certain embodiments, the estimated power for each of the source electrodes **348** is compared to one or more thresholds. Different thresholds can be associated with different sets of sink electrodes **348** (e.g., proximal electrodes, distal electrodes). If the estimated power for a given source electrode **348** is above a predetermined threshold, the GUI **300** can indicate that the threshold is violated. For example, if one or more source electrodes **348** are estimated to receive power above the threshold, the GUI **300** can display a warning message or alarm and/or can display the power estimation icon **362** associated with the violating sink electrode **348** in a different color (e.g., red) than the power estimation icons **362** associated with the other sink electrodes **348**. In FIG. 5, the power estimation icon **362** is shown as being displayed differently than the other power estimation icon **362** to indicate that a threshold has been breached. In certain embodiments, if the power estimated at one of the sink electrodes **348** violates the threshold, the system **100** will not be able to initiate an ablation procedure. For example, so long as one of the estimated power amounts violates the threshold, certain icons on the GUI **300** for initiating energy delivery to the ablation catheter (e.g., the ablation activate/deactivate icon **344**) can be disabled. The disabled ablation icon(s) can be enabled once all of the estimated power amounts are below respective thresholds. As noted above, the power estimates can be updated dynamically as power, etc., is modified by the user.

(42) In certain embodiments, upon selecting the ablation activate/deactivate icon **344** or another icon on the GUI **300**, the source electrodes **346** are automatically assigned an amount of power in accordance with certain goals. In one example, the source electrodes **346** are assigned a power such that an estimated power associated with each of the designated sink electrodes **348** is less than a predetermined threshold (e.g., 10 Watts, 11 Watts). As described above, different thresholds can be applied to different sets of sink electrodes **348** such that estimated power for each of the proximal sink electrodes are compared against a different threshold than the distal sink electrodes. In another example, the source electrodes **346** are assigned a power such that an estimated power associated

with each of the designated sink electrodes **348** such that the estimated power for each of the designated sink electrodes is substantially similar (e.g., within 5%, within 10%) to the estimated power of the other sink electrodes. When the sink electrodes **348** have similar estimated powers, it is more likely that lesions created by the ablation catheter will be more uniform (e.g., lesions with same or similar depth along the lesion).

(43) Once the user sets the desired power for each of the source electrodes **346** (e.g., assisted by viewing the estimated power amounts for the sink electrodes **348**), the user can select the ablation activate/deactivate icon **344**. Selecting the ablation activate/deactivate icon **344** initiates and/or stops energy delivery to the ablation electrodes of the ablation catheter. Once the activate/deactivate icon **344** is pressed to initiate energy delivery, a graphic in the activate/deactivate icon **344** changes (e.g., changes to a stop sign). Further, once the activate/deactivate icon **344** is selected to initiate energy delivery, a signal is transmitted to an RF generator (e.g., the RF generator **114** of FIG. **1**) and/or an RF generator controller (e.g., the RF generator controller **116** of FIG. **1**) to start delivering energy to the selected ablation electrodes of the ablation catheter.

(44) The various methods and functions described above can be carried out by components discussed above, for example, computing devices with one or more controllers implemented using firmware, integrated circuits, and/or software modules that interact with each other or are combined together. The controllers may include memory storing computer-readable instructions/code (e.g., computer program products) for execution by a processor to perform aspects discussed herein.

(45) Various modifications and additions can be made to the exemplary embodiments discussed without departing from the scope of the present invention. For example, while the embodiments described above refer to particular features, the scope of this invention also includes embodiments having different combinations of features and embodiments that do not include all of the described features. Accordingly, the scope of the present invention is intended to embrace all such alternatives, modifications, and variations as fall within the scope of the claims, together with all equivalents thereof.

Claims

1. An ablation system comprising: an energy generator configured to generate ablation energy; an ablation catheter in communication with the energy generator and including a plurality of electrodes; and one or more controllers in communication with the energy generator and configured to: generate a graphical representation of the plurality of electrodes of the ablation catheter for displaying via a graphical user interface (GUI); designate, via the GUI, at least some of the plurality of electrodes to be active electrodes; designate the active electrodes as a source electrode or a sink electrode, wherein at least two or more of the active electrodes being designated as source electrodes and at least two or more of the active electrodes being designated as sink electrodes; assign an amount of power to each of the designated source electrodes; estimate an amount of power associated with each of the designated sink electrodes based at least in part on the assigned amount of power of each of the designated source electrodes and based at least in part on distances between the designated source electrodes and the designated sink electrodes; make a determination that the estimated amount of power associated with at least one of the designated sink electrodes is above a threshold; and disable an ablation-initiation button on the GUI in response to the determination.
2. The ablation system of claim 1, wherein the distances between the designated source electrodes and the designated sink electrodes represent edge-to-edge distances.
3. The ablation system of claim 1, wherein the distances are selected from a table of predetermined distances stored in the one or more controllers computed or configured to be computed by the one or more controllers.

4. The ablation system of claim 1, wherein the estimated amount of power associated with each of the designated sink electrodes is based at least in part on adding each designated source electrode's contribution to the amount of power associated with each of the designated sink electrodes.
5. The ablation system of claim 1, wherein the one or more controllers are configured to: modify the assigned amount of power to at least one of the designated source electrodes; and re-estimate the estimated amount of power associated with each of the designated sink electrodes based at least in part on the modified assigned amount of power of the at least one of the designated source electrodes.
6. The ablation system of claim 5, wherein the one or more controllers are configured to: after re-estimating the estimated amount of power associated with each of the designated sink electrodes, determine that the estimated amount of power associated with at least one of the designated sink electrodes is below the threshold; and enable the ablation-initiation button on the GUI in response to determining that the estimated amount of power associated with the at least one of the designated sink electrodes is below the threshold.
7. The ablation system of claim 1, wherein the ablation catheter includes an expandable member carrying the plurality of electrodes.
8. The ablation system of claim 1, further comprising: a display in communication with the one or more controllers and configured to display the GUI.
9. A computing device for generating and using a graphical user interface (GUI), the computing device comprising: one or more controllers configured to: generate a graphical representation of a plurality of electrodes of an ablation catheter for displaying via the GUI, the plurality of electrodes of the ablation catheter in communication with the one or more controllers; designate, via the GUI, at least some of the plurality of electrodes to be active electrodes; automatically designate the active electrodes as a source electrode or a sink electrode, wherein at least two or more of the active electrodes being designated as source electrodes and at least two or more of the active electrodes being designated as sink electrodes; assign an amount of energy to each of the designated source electrodes; estimate an amount of energy associated with each of the designated sink electrodes based at least in part on the assigned amount of energy of each of the designated source electrodes and based at least in part on distances between the designated source electrodes and the designated sink electrodes; apply the estimated amount of energy to the designated sink electrodes of the ablation catheter; make a determination that the estimated amount of energy associated with at least one of the designated sink electrodes is above a threshold; and disable an ablation-initiation button on the GUI in response to the determination, the ablation-initiation button configured to enable the assigned and estimated amounts of energy provided via the one or more controllers to the plurality of electrodes of the ablation catheter.
10. The computing device of claim 9, wherein the distances between the designated source electrodes and the designated sink electrodes represent edge-to-edge distances.
11. The computing device of claim 9, wherein the distances are selected from a table of predetermined distances stored in the one or more controllers.
12. The computing device of claim 9, wherein the estimated amount of energy associated with each of the designated sink electrodes is based at least in part on adding each designated source electrode's contribution to the amount of energy associated with each of the designated sink electrodes.
13. The computing device of claim 9, wherein the one or more controllers are configured to: designate an additional electrode of the plurality of electrodes as an active electrode or inactive electrode; and re-estimate the estimated amount of energy associated with each of the designated sink electrodes based at least in part on the additional electrode.
14. The computing device of claim 13, wherein the one or more controllers are configured to: after re-estimating the estimated amount of energy associated with each of the designated sink electrodes, determine that the estimated energy associated with at least one of the designated sink

electrodes is below the threshold; and enable the ablation-initiation button on the GUI in response to determining that the estimated amount of energy associated with the at least one of the designated sink electrodes is below the threshold.

15. An ablation system comprising: an energy generator; an ablation catheter in communication with the energy generator and including a plurality of electrodes; and one or more controllers in communication with the energy generator and configured to: generate a graphical representation of the plurality of electrodes of the ablation catheter for displaying via a graphical user interface (GUI); designate, via the GUI, at least some of the plurality of electrodes to be active electrodes; designate the active electrodes as a source electrode or a sink electrode, wherein at least two or more of the active electrodes being designated as source electrodes and at least two or more of the active electrodes being designated as sink electrodes; assign an amount of power to each of the designated source electrodes; estimate an amount of power associated with each of the designated sink electrodes based at least in part on the assigned amount of power of each of the designated source electrodes and distances between the designated source electrodes and the designated sink electrodes; and apply the estimated amount of power to via the energy generator to the ablation catheter.

16. The ablation system of claim 15, wherein the distances between the designated source electrodes and the designated sink electrodes represent edge-to-edge distances.

17. The ablation system of claim 15, wherein the estimated amount of power associated with each of the designated sink electrodes is based at least in part on adding each designated source electrode's contribution to the amount of power associated with each of the designated sink electrodes.

18. The ablation system of claim 15, wherein the one or more controllers are configured to: modify the assigned amount of power to at least one of the designated source electrodes; and re-estimate the estimated amount of power associated with each of the designated sink electrodes based at least in part on the modified assigned amount of power of the at least one of the designated source electrodes.
